CareVoice AI

Governed Voice + RAG for Senior & Patient Care

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Category: Azure AI Governed RAG

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AI wellness companion for seniors living alone • Grounded in data • Safe by default • Fully observable

The Problem

14M Seniors Live Alone — No Daily Check-ins

Seniors living alone go days without a meaningful conversation. Declining nutrition, isolation, and emotional distress go undetected until a crisis. Families and caregivers have no scalable way to monitor wellness daily.

LLMs Hallucinate in Healthcare

Vanilla LLMs generate plausible but ungrounded responses. For a senior living alone, a hallucinated health recommendation can cause real harm with no one there to catch it.

No Safety Guardrails

Most AI voice systems have zero content safety enforcement. Medical advice slips through. PHI gets exposed. For vulnerable populations like seniors, this is unacceptable.

Black Box Conversations

When something goes wrong in a patient or senior care call, there's no way to trace what happened, why the AI said what it said, or whether safety checks ran.

CareVoice AI: The Solution

A governed, voice-enabled RAG system that conducts daily wellness check-in calls for seniors living alone and patients in care — with grounded responses, safety enforcement, and full observability.

Voice-First

Twilio + ConversationRelay for real-time streaming calls

RAG-Grounded

Azure AI Search hybrid retrieval ensures every response is data-backed

Safety Enforced

Azure Content Safety validates every single LLM output

Fully Observable

OpenTelemetry + App Insights traces every span end-to-end

LLM Evaluated

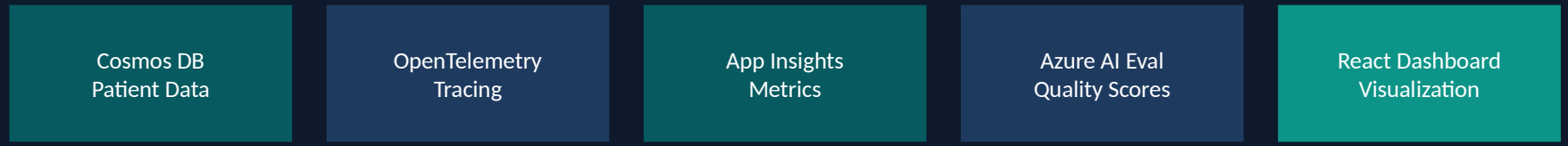
Azure AI Evaluation scores groundedness, relevance, coherence, fluency

Patient Context

Cosmos DB stores patient profiles and session history

End-to-End Architecture

Supporting Services



System Flow

1. Twilio call → 2. FastAPI WebSocket → 3. Speech-to-text → 4. End-of-call detection → 5. Embedding (Azure OpenAI) → 6. Hybrid retrieval (AI Search)
7. Context → LLM (GPT-4o-mini) → 8. Content Safety check → 9. Twilio voice response → 10. OpenTelemetry logging → 11. Post-call LLM evaluation

Azure Services — Breadth & Depth

7 Azure services working together in a single governed pipeline

Azure OpenAI Service

Core LLM + Embeddings

GPT-4o-mini for conversational responses + embedding generation for RAG retrieval

Azure AI Search

RAG Retrieval

Hybrid retrieval (keyword + semantic) over patient wellness knowledge base

Azure AI Content Safety

Safety Layer

Real-time content moderation on every LLM output — blocks harmful/ungrounded content

Azure Cosmos DB

Data Store

Patient profiles, session history, wellness scores — globally distributed NoSQL

Azure Application Insights

Observability

Receives OpenTelemetry traces and metrics — full call lifecycle visibility

Azure AI Evaluation

Quality Metrics

Post-call quality scoring: groundedness, relevance, coherence, fluency

Azure App Service

Compute / Hosting

Hosts FastAPI backend with auto-scaling for production-grade deployment

RAG Pipeline — How It Works

1. **Patient Utterance**
   Speech captured via Twilio ConversationRelay WebSocket stream
2. **Embedding Generation**
   Azure OpenAI creates vector embedding of the patient query
3. **Hybrid Retrieval**
   Azure AI Search runs keyword + semantic search over wellness knowledge base
4. **Context Assembly**
   Top retrieved documents assembled with patient context from Cosmos DB
5. **LLM Generation**
   GPT-4o-mini generates grounded response using assembled context
6. **Safety Validation**
   Azure Content Safety scans output — blocks harmful, ungrounded, or PHI-containing responses
7. **Voice Delivery**
   Safe response streamed back to patient via Twilio voice

RAG Governance Features

- Hybrid search (keyword + semantic vectors)
- Patient-specific context injection from Cosmos DB
- Medical advice restriction guardrail
- PHI awareness — flags protected health info
- Groundedness check on every response
- Citation tracking for traceability
- Post-call evaluation: groundedness, relevance, coherence, fluency scores

Responsible AI — Safety at Every Layer

25% of judging criteria — here's how CareVoice AI delivers

Azure Content Safety on Every Turn

Every single LLM response is scanned before it reaches the patient. Harmful, ungrounded, or inappropriate content is blocked in real time.

Medical Advice Restriction

System prompt guardrails explicitly prevent diagnosis, prescriptions, or treatment recommendations. CareVoice checks wellness — it doesn't practice medicine.

PHI Awareness

Protected Health Information is flagged. The system is designed to avoid exposing or generating PHI in uncontrolled contexts.

Groundedness Enforcement

RAG pipeline ensures responses are backed by retrieved data, not hallucinated. Azure AI Evaluation scores groundedness post-call.

Full Auditability

OpenTelemetry traces capture every step: call start, utterance handling, safety check, Cosmos lookup, evaluation run. Nothing is a black box.

LLM Quality Evaluation

Azure AI Evaluation computes groundedness, relevance, coherence, and fluency scores. Enables continuous quality monitoring.

Observability — Everything is Traceable

Tracked OpenTelemetry Spans

`carevoice.start_call`

Call initiation, patient lookup, session creation

`carevoice.handle_utterance`

Full utterance processing: embed → retrieve → generate → safety

`content_safety_check`

Azure Content Safety scan with pass/fail result

`cosmos_lookup`

Patient profile and history retrieval from Cosmos DB

`evaluation_run`

Post-call Azure AI Evaluation: groundedness, relevance, coherence, fluency

Observability Stack

OpenTelemetry SDK

Instrument code with custom spans and attributes. OTLP exporter sends data to collectors.

Azure Application Insights

Receives traces and metrics in Azure Portal. Query with KQL, build dashboards, set alerts.

Aspire Dashboard

Local dev observability — see traces in real time during development and testing.

React Dashboard

Custom frontend with Recharts showing wellness scores, RAG metrics, pipeline status, and traces.

Microsoft Agent Framework — Pipeline Design

Designed for MAF orchestration — currently implemented as deterministic procedural pipeline

Triage Agent → **Retrieval Agent** → **Response Agent** → **Safety Agent** → **Alert Agent**

Triage Agent	Retrieval Agent	Response Agent	Safety Agent	Alert Agent
Classifies intent: wellness check, emergency, general	Runs hybrid search on Azure AI Search, assembles context	GPT-4o-mini generates grounded response from RAG context	Azure Content Safety validates output, blocks if needed	Detects escalation triggers, flags for clinician review

Current Implementation

- Procedural pipeline in FastAPI
- Each stage runs sequentially in workflows/
- Deterministic control — explicit logic, no magic
- Production-ready and fully testable

Future: Full MAF Orchestration

- Multi-agent routing with dynamic handoff
- Agent memory for personalization across calls
- Clinician escalation agent for critical flags
- Full Microsoft Agent Framework SDK integration

Dashboard — Monitoring & Evaluation

Wellness Monitoring

Tracks patient nutrition, physical, emotional, and social wellness scores. Radar chart visualization with live call transcript display.

RAG Governance

Displays groundedness, relevance, coherence, and fluency scores from Azure AI Evaluation. Citation tracking and safety check results per conversation.

Pipeline View

Visualizes the Triage → Retrieval → Response → Safety → Alert flow. Shows stage-by-stage status and processing outcomes.

Observability

OpenTelemetry trace viewer showing spans, durations, and attributes. Performance metrics and safety evaluation status for each call.

Innovation & Performance

Innovation (25%)

- Voice-first RAG — most RAG systems are text-only; CareVoice operates in real-time voice, perfect for seniors who can't use apps
- Dual use case — daily wellness companion for seniors living alone + patient care follow-up for healthcare providers
- Governed pipeline — every response is safety-checked before reaching the senior or patient
- Intelligent call termination — detects "bye", "thank you", fuzzy speech like "buh"
- Full evaluation loop — LLM outputs are scored post-call, enabling continuous quality improvement

Performance (25%)

Real-time

Voice streaming via Twilio ConversationRelay WebSocket — no batch delays

Async

FastAPI with async/httpx — non-blocking I/O for concurrent call handling

Hybrid

Azure AI Search keyword + semantic retrieval for high-precision, high-recall RAG

Scalable

Cosmos DB globally distributed + Azure App Service auto-scaling for production load

How CareVoice AI Scores on Every Criteria

Performance — 25%

- Real-time voice streaming (no batch latency)
- Async FastAPI with httpx for concurrent calls
- Hybrid retrieval for precision + recall
- Cosmos DB + App Service auto-scaling

Innovation — 25%

- Voice-first RAG (not text-only)
- Governed pipeline with safety on every turn
- Intelligent fuzzy call termination
- Full post-call LLM evaluation loop

Azure Services Breadth — 25%

- 7 Azure services: OpenAI, AI Search, Content Safety, Cosmos DB, App Insights, AI Eval, App Service
- Each service plays a distinct, critical role
- Deeply integrated — not bolted on

Responsible AI — 25%

- Content Safety on every LLM output
- Medical advice restriction + PHI awareness
- Groundedness enforcement via RAG
- Full auditability with OpenTelemetry

What You'll See in Azure Portal

Show these in your video walkthrough to demonstrate the live system

Application Insights → Transaction Search

Show the OpenTelemetry traces arriving in real time. Filter by "carevoice.start_call" or "content_safety_check". Click into a trace to show the full span waterfall — this proves observability is live, not mocked.

Application Insights → Live Metrics

While on the phone call, show the Live Metrics stream. The audience sees requests and traces flowing in real time as you speak to CareVoice AI.

Azure AI Search → Index

Show the search index with the wellness knowledge base documents. Run a sample query to demonstrate hybrid retrieval returning ranked results.

Azure Content Safety → Dashboard

Show the Content Safety resource and its monitoring. Demonstrate that it's integrated by pointing to the API calls in App Insights traces.

Cosmos DB → Data Explorer

Show patient records and session data. Query a recent call session to show stored context — proves the data pipeline is real.

Complete Technology Stack

AI & RAG	Azure OpenAI (GPT-4o-mini) • Azure OpenAI Embeddings • Azure AI Search (hybrid)
Safety	Azure AI Content Safety • Medical advice restriction • PHI awareness guardrails
Data	Azure Cosmos DB (patient profiles + sessions) • Azure AI Search (knowledge base)
Voice	Twilio Voice API • Twilio ConversationRelay (WebSocket streaming)
Backend	Python • FastAPI (async) • httpx • WebSocket handlers
Observability	OpenTelemetry (OTLP) • Azure Application Insights • Aspire Dashboard
Evaluation	Azure AI Evaluation SDK — groundedness, relevance, coherence, fluency
Frontend	React • Vite • TypeScript • Recharts
Deployment	Azure App Service • Docker • docker-compose.yml

Try It Yourself — Live Demo

1-866-505-5395

Call now — the line is live

How to experience CareVoice AI:

- Dial the number above from any phone
- When prompted, say "Margaret" — this loads a full patient profile with wellness history
- Have a natural conversation — ask about nutrition, how you're feeling, or daily activity
- Say "bye" or "thank you" to end — intelligent call termination handles the rest

Live Demo & Dashboard

Live Deployment

CareVoice AI Live Deployment
thankful-cliff-0f698610f.1.azurestaticapps.net

Video Demo

Watch Demo on Google Drive

Live Dashboard

During the call, see real-time updates. After the call ends, review results across dashboards:

- Wellness Monitor
- RAG Governance
- Agent Pipeline View
- Observability

CareVoice AI

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Thank You

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